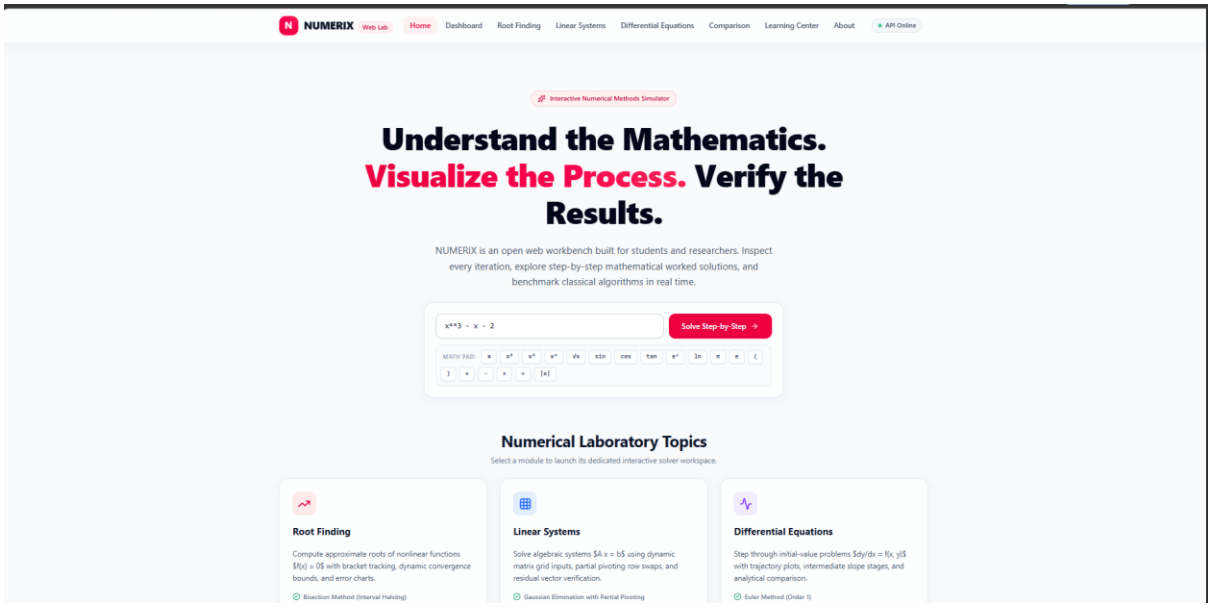
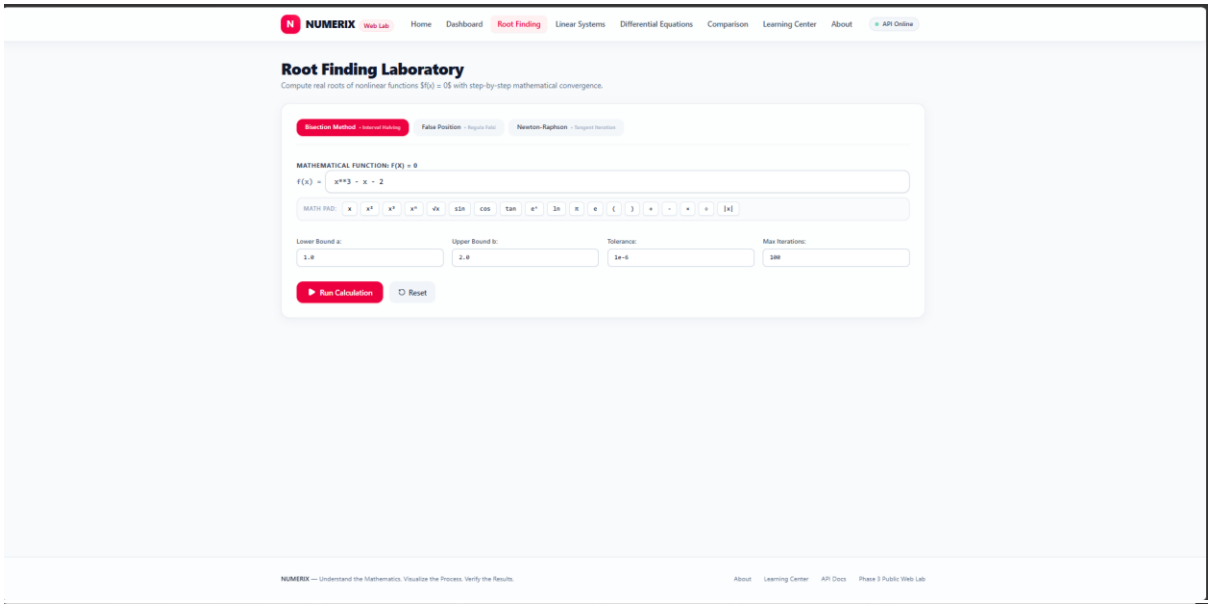
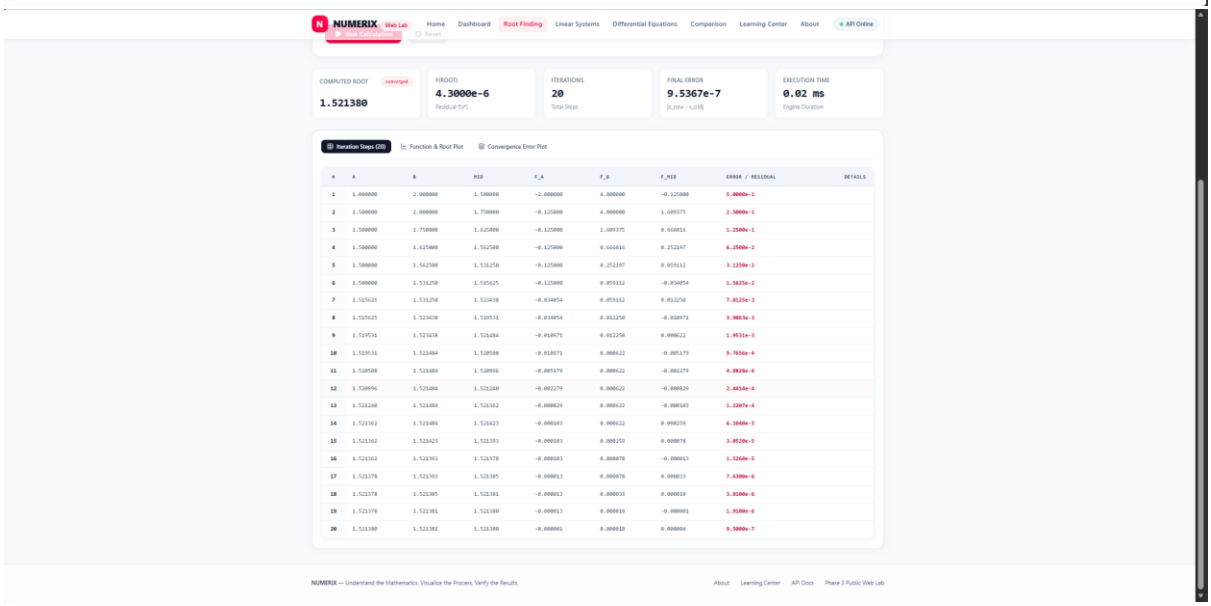
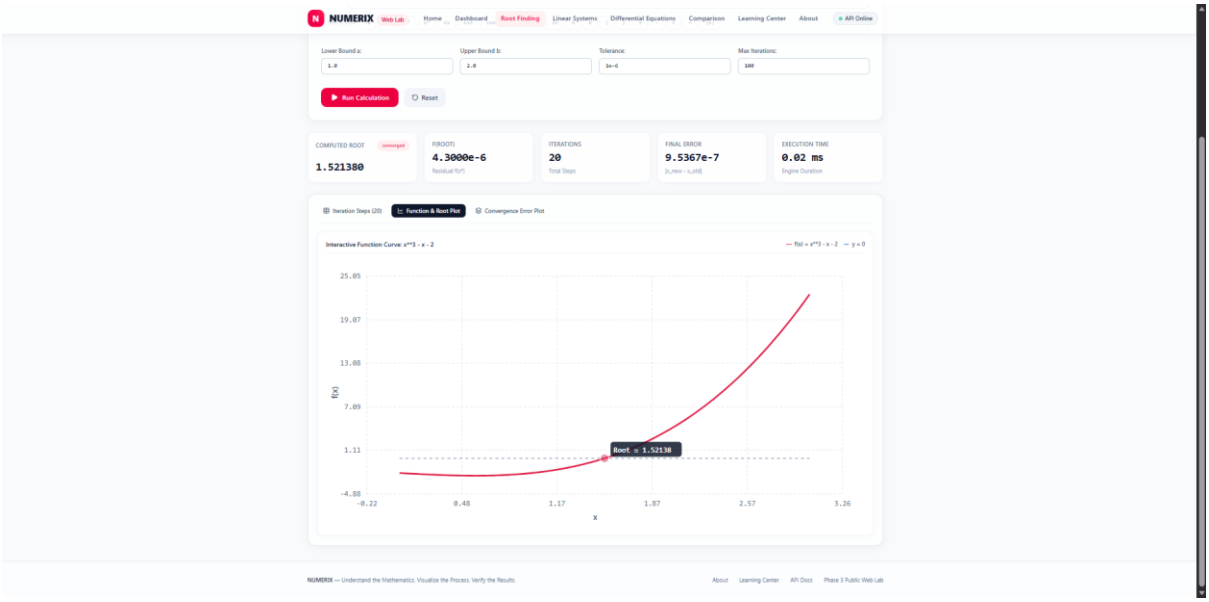
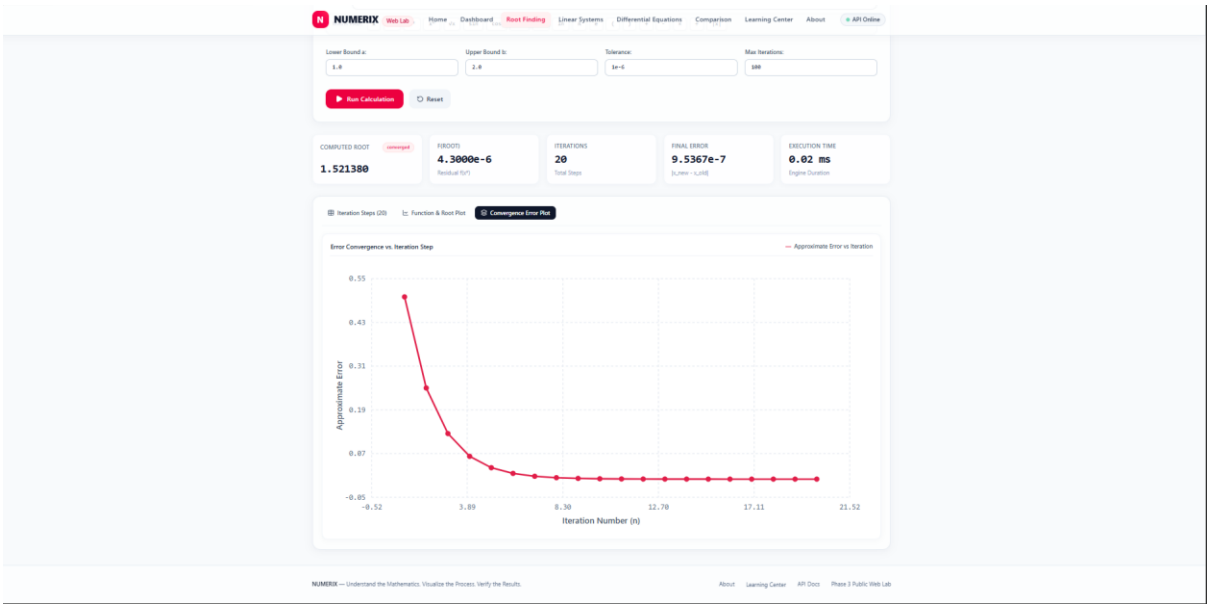

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Differential Equations
Comparison
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Interactive Numerical Methods Simulator
Understand the Mathematics.
Visualize the Process. Verify the Results.
NUMERIX is an open web workbench built for students and researchers. Inspect every iteration, explore step-by-step mathematical worked solutions, and benchmark classical algorithms in real time.
x**3 - x - 2
Solve Step-by-Step
MATH PAD:
Numerical Laboratory Topics
Select a module to launch its dedicated interactive solver workspace.
Root Finding
Compute approximate roots of nonlinear functions $f(x) = 0$ with bracket tracking, dynamic convergence bounds, and error charts.
Bisection Method (Interval Halving)
Linear Systems
Solve algebraic systems $A x = b$ using dynamic matrix grid inputs, partial pivoting row swaps, and residual vector verification.
Gaussian Elimination with Partial Pivoting
Differential Equations
Step through initial-value problems $dy/dx = f(x, y)$ with trajectory plots, intermediate slope stages, and analytical comparison.
Euler Method (Order 1)


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Root Finding Laboratory
Compute real roots of nonlinear functions $f(x) = 0$ with step-by-step mathematical convergence.
Bisection Method - Interval Halving
False Position - Regula Falsi
Newton-Raphson - Tangent Iteration
MATHEMATICAL FUNCTION: F(X) = 0
f(x) =
x**3 - x - 2
MATH PAD:
Lower Bound a:
1.0
Upper Bound b:
2.0
Tolerance:
1e-6
Max Iterations:
100
Run Calculation
Reset
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COMPUTED ROOT
converged
1.521380
RESIDUAL f(r)
4.3000e-6
ITERATIONS
20
Total Steps
FINAL ERROR
9.5367e-7
EXECUTION TIME
0.02 ms
Engine Duration
Iteration Steps (20)
Function & Root Plot
Convergence Error Plot
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1.521380
f(ROOT)
4.3000e-6
Residual f(x*)
ITERATIONS
20
Total Steps
FINAL ERROR
9.5367e-7
|x_new - x_old|
EXECUTION TIME
0.02 ms
Engine Duration
Iteration Steps (20)
Function & Root Plot
Convergence Error Plot
Interactive Function Curve: x**3 - x - 2
f(x) = x**3 - x - 2
y = 0
25.05
19.07
13.08
7.09
1.11
-4.88
f(x)
Root: x = 1.52138
-0.22
0.48
1.17
1.87
2.57
3.26
x
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Residual f(x*)
ITERATIONS
20
Total Steps
FINAL ERROR
9.5367e-7
|x_new - x_old|
EXECUTION TIME
0.02 ms
Engine Duration
Iteration Steps (20)
Function & Root Plot
Convergence Error Plot
Error Convergence vs. Iteration Step
Approximate Error vs Iteration
0.55
0.43
0.31
0.19
0.07
-0.05
Approximate Error
-0.52
3.89
8.30
12.70
17.11
21.52
Iteration Number (n)
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